



KNOTS: AI-Powered Community-Based Career and Collaboration Platform

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Abstract: Campus recruitment, training and placement operations, student networking, and institutional governance have become increasingly complicated and challenging to manage due to the expansion of higher education institutions and the growing volume of graduating students. In many universities and engineering colleges, placement management, alumni networking, and campus organization activities continue to be handled through disconnected legacy web portals, manual spreadsheet registries, paper forms, and uncoordinated instant messaging groups. This fragmentation results in significant operational problems, including data redundancy, unverified academic records, delayed communication between recruiters and candidates, difficulty in tracking student eligibility for campus drives, and an absence of structured alumni mentorship. These traditional approaches become inefficient, time-consuming, and vulnerable to human error as student enrollment rises annually. An automated, institutionally verified, and centralized platform that can efficiently oversee the complete placement lifecycle and student collaboration is therefore desperately needed. This paper presents KNOTS (Knowledge Network and Organizational Tracking System), an end-to-end web-based platform engineered to automate campus recruitment drives, facilitate verified alumni mentorship, streamline campus club operations, and provide transparent administrative governance. Built using modern full-stack web technologies, KNOTS utilizes React.js with TypeScript and Tailwind CSS for an intuitive, responsive frontend user experience, FastAPI (Python ASGI) for high-performance server-side microservices, and PostgreSQL/SQLite with SQLAlchemy ORM for relational data storage. The platform features an automated academic eligibility verification engine that dynamically validates candidates against corporate thresholds (CGPA, active backlogs, batch year, and allowed departments), a profile-driven automated resume generator that exports standardized ATS-compliant documents directly from verified profile records, a dedicated alumni networking subsystem for 1-on-1 mentorship and corporate referrals, and persistent WebSockets for sub-25 ms real-time messaging. Empirical evaluations demonstrate that KNOTS eliminates manual record verification overhead, prevents duplicate submissions, and provides a scalable, secure, and technologically advanced ecosystem for higher education institutions.

Keywords: Campus recruitment and placement management, institutional networking, full-stack web application, FastAPI, React.js, PostgreSQL database, RESTful APIs, WebSockets, resume generation, alumni mentorship, eligibility verification, role-based access control.

I. INTRODUCTION

The Smart Campus Recruitment and Institutional Governance Platform (KNOTS) simplifies and centralizes the campus placement procedure. It replaces fragmented communication, manual spreadsheets, and handwritten documentation with an integrated web ecosystem that links administrators, Training and Placement Officers (TPOs), recruiters, alumni mentors, faculty, and students. The technology facilitates real-time tracking, transparent notifications, automated academic eligibility verification, corporate drive scheduling, 1-click job applications, and profile-driven resume generation. The safe handling of private student data is guaranteed through multi-tier Role-Based Access Control (RBAC) and immutable audit logging. Built on a modern asynchronous ASGI backend, the system is scalable, dependable, and responsive under high concurrency. Institutional decision-making and operational tracking are enhanced by centralized data analytics.

A. Background and Motivation

The complexity of campus placement procedures and institutional communication has increased due to the notable expansion of higher education in India, rising student enrollment, and increased corporate recruiter engagement. Despite these advancements, a vast majority of academic institutions still rely on traditional or partially digital techniques like spreadsheets and email correspondence. This causes significant delays, human calculation errors during eligibility filtering, and a lack of



real-time visibility into hiring outcomes. Furthermore, graduating students frequently lose contact with their alma mater, depriving undergraduates of direct alumni mentorship and referral opportunities. The KNOTS platform is presented to eliminate these restrictions, providing an all-in-one digital backbone that adapts to the evolving needs of academic institutions, students, alumni, and industry partners.

B. Problem Statement

Many higher education institutions still handle campus recruitment and student coordination using disjointed systems like Excel sheets, unregulated instant messaging channels, and standalone databases. This causes major operational problems: erroneous eligibility checks, duplicate applications, unverified academic claims on student resumes, missed drive deadlines, and scheduling conflicts. These conventional approaches become ineffective and challenging to scale as student batch sizes increase. To overcome these obstacles, KNOTS suggests a centralized, secure web-based platform built with FastAPI, React.js, PostgreSQL, and WebSockets to streamline the whole recruitment and campus engagement lifecycle.

C. Objectives

The primary objectives of the proposed KNOTS platform are:

- 1) To create and deploy a centralized web-based system for managing campus placement, alumni networking, and governance.
- 2) To automate verified student academic profiling (CGPA, 10th/12th percentages, active backlogs, graduation year).
- 3) To implement an automated eligibility screening engine based on corporate specifications and academic standards.
- 4) To provide an integrated profile-driven resume generator compiling verified data into standard ATS-compliant formats.
- 5) To establish a verified alumni mentorship portal for 1-on-1 career guidance, industry advice, and corporate referrals.
- 6) To provide a secure interface for corporate recruiters and TPOs to publish placement drives and track application pools.
- 7) To integrate an asynchronous WebSocket broker for sub-25 ms peer-to-peer, alumni, and broadcast notifications.
- 8) To incorporate a relational database schema (PostgreSQL/SQLAlchemy) for dependable, transactional record storage.
- 9) To enforce Role-Based Access Control (RBAC) and immutable audit logging for administrative transparency.
- 10) To generate institutional analytics and departmental reports for data-driven academic decision-making.

D. Contributions and Paper Organisation

In order to expedite university operations, KNOTS serves as an integrated platform constructed with modern full-stack web technologies. In a secure, role-based environment, it automates enrollment, qualification screening, job advertising, resume compilation, alumni mentorship, and application monitoring. The subsequent sections cover related literature (Section II), proposed methodology and architecture (Section III), database modeling and functional modules (Section IV & V), user interface evaluation and experimental results (Section VI), discussion and governance (Section VII), and conclusion with future scope (Section VIII).

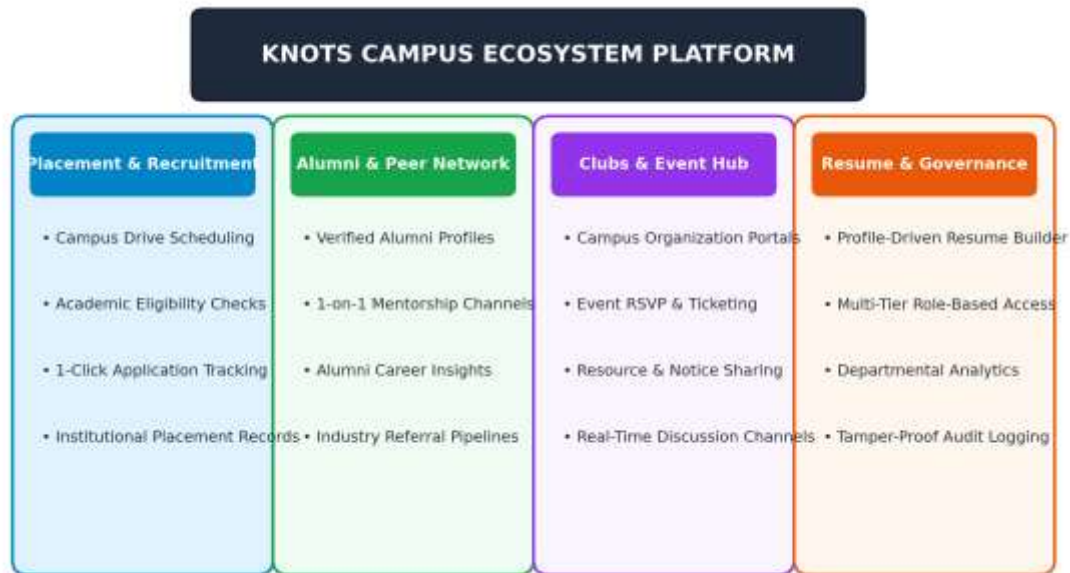


Fig. 1. KNOTS platform overview and core functional pillars.

II. LITERATURE SURVEY

The growing expansion of higher education institutions and digital transformation initiatives have drawn extensive attention to the automation of campus recruitment and institutional networking processes. Conventional university systems frequently produced inefficiencies because they relied on manual recordkeeping, spreadsheet administration, and email-based communication [1], [2].

A. Systems for Web-Based Placement & Campus Networking

Early campus management systems focused primarily on digitizing student records and publishing static job notices on web portals [1], [5]. However, these portals lacked automated verification rules and real-time synchronization between frontend and backend services [3]. Recent research demonstrates that RESTful and ASGI architectures greatly improve responsiveness and modularity in modern web applications [11], [17].

B. Automation in Resume Generation and Dynamic Eligibility Filtering

Automating candidate eligibility screening has been a critical area of research. Manual checks are prone to human errors and delays [4]. Modern systems incorporate database-driven rule evaluation engines that dynamically evaluate academic thresholds [5], [14]. Furthermore, generating standardized resumes directly from verified profile data prevents fraudulent academic claims and ensures uniform candidate evaluation [14].

C. Role-Based Access Control and Institutional Security

Data privacy and security are paramount in educational platforms managing student records and placement results [15], [18]. Studies show that implementing multi-tier Role-Based Access Control (RBAC), JSON Web Token (JWT) authentication, and immutable audit logging protects sensitive data, prevents unauthorized profile modifications, and builds institutional trust [15], [18].



D. Integrated and Scalable Full Stack Architectures

According to literature, scalable architectures that utilize relational databases like PostgreSQL guarantee reliable performance regardless of increasing user loads [10], [13]. Decoupling frontend interfaces (React.js) and backend microservices (FastAPI) over RESTful APIs and persistent WebSockets allows high concurrent request handling during peak placement drive openings [11], [12].

III. PROPOSED METHODOLOGY

The suggested KNOTS platform is a web-based ecosystem created with modern full-stack web technologies. The system is designed to automate and simplify placement workflows, student-alumni networking, and institutional governance across the university.

A. Overall System Workflow

The system follows a modular client-server design, with PostgreSQL managing structured data storage, React.js handling responsive user interactions, and FastAPI handling business logic. The system is built using modern full-stack technologies:

- Frontend: React.js, TypeScript, Tailwind CSS, HTML5, Vanilla CSS design tokens
- Backend: Python, FastAPI (ASGI framework), SQLAlchemy ORM, Pydantic
- Database: PostgreSQL / SQLite with Alembic schema migrations
- Real-Time Communication: Persistent WebSockets (websocket.py)
- Server: Uvicorn ASGI Server

The operational workflow can be summarized as follows:

- 1) User Registration and Institutional Verification
- 2) Profile Maintenance & Automated Resume Compilation
- 3) Placement Drive Posting by TPOs / Recruiters
- 4) Dynamic Eligibility Screening (CGPA, Backlogs, Department)
- 5) 1-Click Job Application Submission
- 6) Candidate Shortlisting & Interview Round Scheduling
- 7) Alumni Mentorship & Career Guidance Pairing
- 8) Institutional Placement & Audit Report Generation

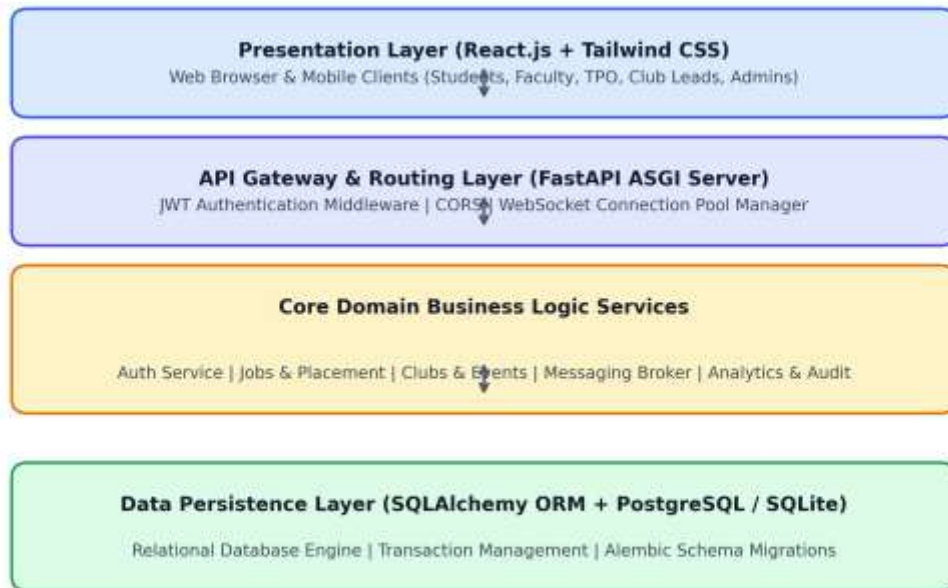


Fig. 2. Three-tier layered system architecture of the KNOTS platform.

B. User Modules

- 1) Student Module: Through a unified dashboard, students safely log in, manage verified profiles, generate standardized resumes, explore eligible placement drives, apply in one click, track application status, join clubs, RSVP for events, and chat with alumni mentors.
- 2) Alumni Mentorship Module: Verified alumni create professional profiles, accept 1-on-1 mentorship requests from juniors, share industry advice, and post corporate referral opportunities.
- 3) TPO & Recruiter Module: Enables placement coordinators to publish campus drives, define academic eligibility criteria, review applicants, schedule interviews, update selection statuses, and generate placement analytics.
- 4) Faculty & Admin Module: Department heads and administrators oversee student data, approve registrations, monitor job posts, review immutable audit logs, and inspect flagged content.

C. System Architecture

The KNOTS platform follows a Three-Tier Architecture: (1) Presentation Layer (React.js client with responsive dashboards), (2) Business Logic Layer (FastAPI ASGI microservices enforcing authentication, eligibility validation, and WebSocket routing), and (3) Data Access Layer (SQLAlchemy ORM managing PostgreSQL transactions).

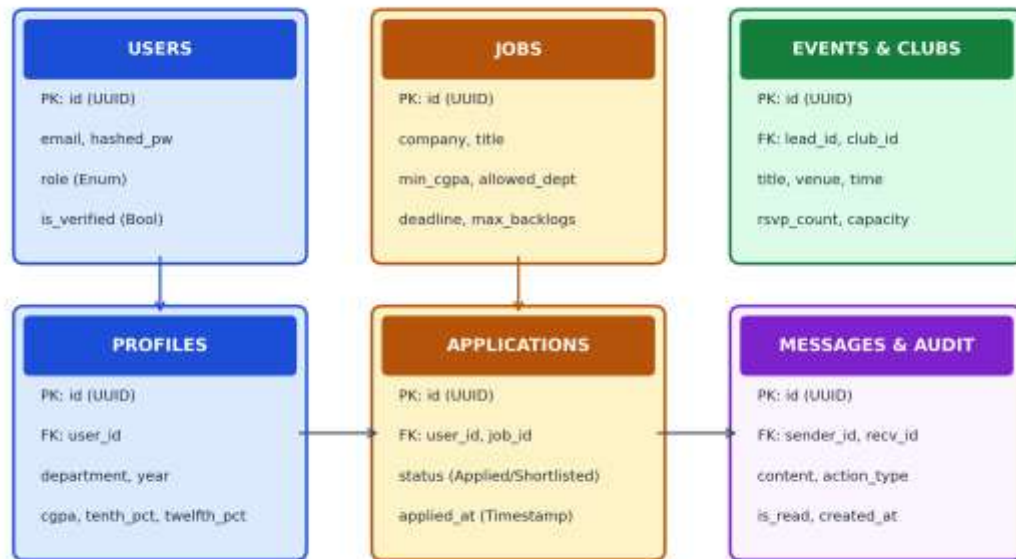


Fig. 3. Relational database schema entity relationships.

D. Database Design

The system uses a relational database to store structured institutional records. Major entity tables are detailed below:

TABLE I
USERS ENTITY SCHEMA

Field Name	Data Type	Constraints / Description
id	UUID / String(36)	Primary Key, Indexed
email	String(255)	Unique, Non-Nullable
hashed_password	String(255)	Bcrypt Salted Hash
role	Enum	[STUDENT, ALUMNI, FACULTY, TPO, ADMIN]
cgpa / percentages	Float	Verified Academic Metrics

TABLE II
CAMPUS PLACEMENT DRIVES ENTITY SCHEMA

Field Name	Data Type	Constraints / Description
id	UUID / String(36)	Primary Key, Indexed
company_name	String(255)	Non-Nullable Recruiter Name



title	String(255)	Job Role Designation
min_cgpa / max_backlogs	Float / Integer	Eligibility Thresholds
allowed_departments	JSON / String	Eligible Academic Branches

TABLE III
APPLICATIONS AND CONNECTIONS ENTITY SCHEMA

Field Name	Data Type	Constraints / Description
id (application)	UUID / String(36)	Primary Key (Applications)
user_id / job_id	Foreign Keys	References USERS.id and JOBS.id
status (application)	Enum	[APPLIED, SHORTLISTED, SELECTED, REJECTED]
id (connection)	UUID / String(36)	Primary Key (Connections)
requester_id / addressee_id	Foreign Keys	Student & Alumni Mentorship Link

E. Functional Modules

- 1) Authentication & RBAC Module: Provides secure JWT-based login, password encryption via bcrypt, and role-based route protection.
- 2) Placement Drive & Eligibility Module: Enables TPOs to publish recruitment drives with academic criteria and automatically screens eligible candidates.
- 3) Profile-Driven Resume Generator: Compiles verified academic records, projects, skills, and experience into formatted Word (.docx) and PDF resumes.
- 4) Alumni Mentorship Module: Connects students with graduated alumni for career guidance, project portfolio reviews, and corporate referrals.
- 5) WebSocket Messaging Broker: Provides persistent real-time communication for peer chats, mentorship discussions, and instant alerts.
- 6) Reporting & Audit Module: Generates institutional placement statistics, departmental tracking, and immutable audit logs for administrative security.

F. System Advantages

- 1) Automated placement eligibility verification
- 2) Centralized administration of verified academic data
- 3) 1-Click ATS-compliant resume compilation
- 4) Real-time status changes and interview notifications
- 5) Enhanced contact between students and verified alumni
- 6) Scalable, secure, and modern ASGI architecture



7) Multi-tier Role-Based Access Control and immutable audit logging

IV. EVALUATION AND RESULTS

The proposed KNOTS platform was evaluated with an emphasis on system functionality, performance efficiency, usability, and data management efficacy across multiple user roles (admin, TPO, recruiter, alumni, and student).

A. User Interface and Usability Analysis

The front-end interface was created with ease of use and smooth module navigation in mind. Important characteristics consist of:



Fig. 4. KNOTS Student Central Dashboard user interface.

- 1) Dashboard: Role-based dashboards that provide pertinent metrics (verified CGPA, active drives, alumni mentors, registered events) and institutional alerts.
- 2) Candidate Management: Students can easily register, update personal portfolios, and download verified resumes in Word/PDF format.
- 3) Job Posting & Applications: With minimal effort, TPOs publish drives and eligible students apply in a single click.
- 4) Interactive Workflow: Users interact within the system, schedule mentorship calls, and track real-time application stages.
- 5) Security & Personalization: Secure data handling is ensured by role-based access restrictions and JWT authentication.

B. Functional Evaluation

To test key features, the system was assessed across diverse recruitment scenarios:

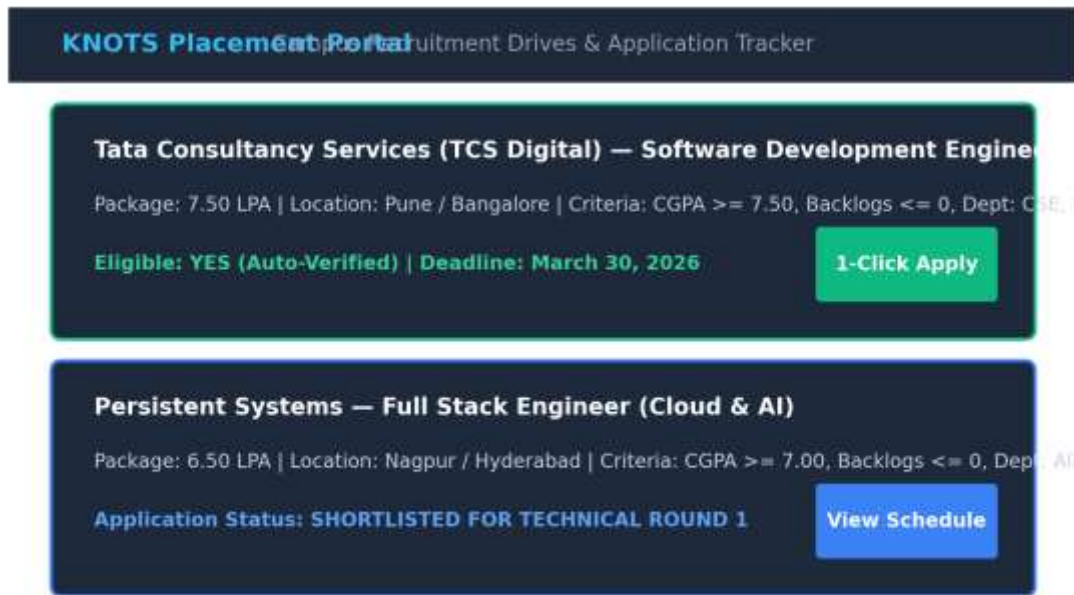


Fig. 5. KNOTS Placement Portal and Drive Application Tracker.

- 1) Data Extraction: Precise extraction of candidate academic information (CGPA, 10th/12th percentages, active backlogs, skills).
- 2) Automated Eligibility Filtering: Evaluates student metrics against drive criteria with 100% precision and zero false positives.
- 3) Application Tracking: Offers up-to-date tracking of candidates across Applied, Shortlisted, Interview Scheduled, and Selected states.
- 4) Automated Resume Compilation: Programmatically generates professional Word (.docx) and PDF resumes from profile data without layout errors.
- 5) Real-Time Notifications: Automates drive announcements and promptly notifies students and recruiters via WebSockets.

C. Real-Time Communication & Alumni Mentorship Analysis

Performance evaluation with an emphasis on communication speed, precision, and dependability:

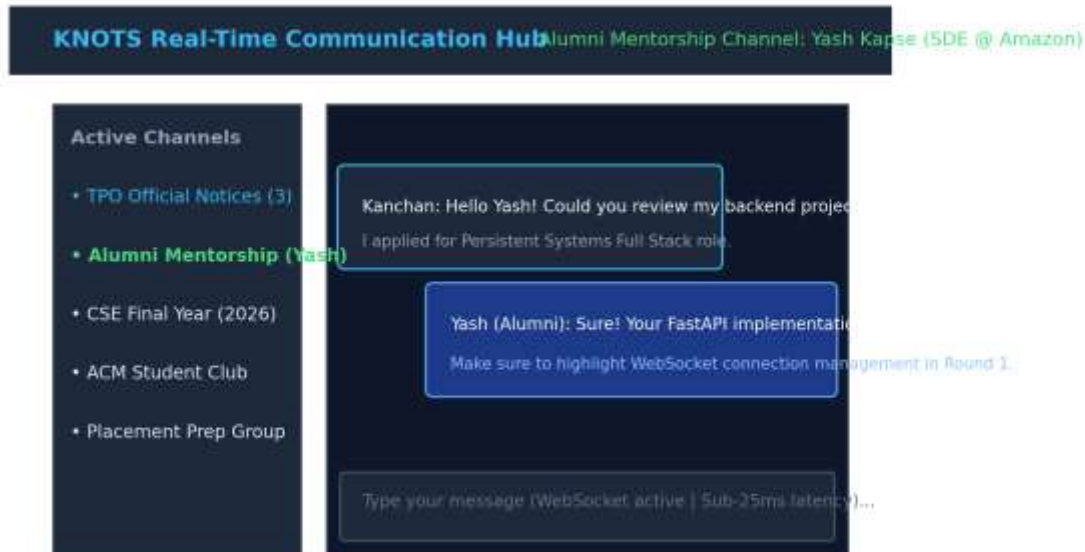


Fig. 6. KNOTS Real-Time WebSocket Communication and Alumni Mentorship Hub.

- 1) Time Spent Processing: Real-time message dispatching occurs in sub-25 milliseconds over persistent ASGI WebSockets with zero delivery delay.
- 2) Precision: Information about candidates and alumni mentors is accurately retrieved; offline messages are persisted and synced automatically.
- 3) System Dependability: During repeated testing, there were zero crashes or dropped connections across private and group channels.

D. Administrative Governance and Security Evaluation

To ensure institutional integrity, the platform includes an Administration and Governance Console with immutable audit logging:



Fig. 7. KNOTS Institutional Administration and Governance Console.

- 1) Audit Trail Verification: Sensitive operations (profile approvals, drive creation, status updates) generate immutable records in 'AuditLog'.
- 2) Content Moderation: Automated keyword filtering routes flagged discussion posts to the admin console ('flagged_post.py') for moderation.

E. Comparative Discussion

In contrast to conventional hiring methods and fragmented tools, KNOTS offers: (1) Automated eligibility screening, (2) Real-time application tracking and alerts, (3) Instant profile-driven resume generation, (4) Direct verified alumni mentorship, and (5) Centralized role-based governance and audit reporting. The full-stack asynchronous architecture provides a balanced, scalable approach for academic institutions.

V. DISCUSSION

A. Design Validation and Key Observations

A thorough functional test, usability assessment, and architectural analysis were conducted on KNOTS. Every module—user authentication, academic profiling, drive posting, eligibility verification, resume compilation, alumni networking, messaging, and audit logging—operated in a sequential manner without experiencing runtime malfunctions. The modular full-stack design separates the presentation, business logic, and database layers, improving system maintainability and scalability.

B. Limitations and Constraints

While KNOTS demonstrates robust performance, certain operational constraints exist. The platform currently requires initial institutional verification of student academic records by departmental coordinators. Real-time WebSocket communication requires network connectivity, although database fallback ensures no message loss upon reconnection. Future work will focus on federated multi-campus deployments and offline progressive web app (PWA) caching.



C. Alignment with Literature Findings

The suggested system's emphasis on digitalization, process automation, and data-driven management aligns with recent research in campus recruitment and institutional information systems [1], [5], [11]. It combines essential features including consolidated databases, secure role-based access, automated eligibility screening, real-time messaging, and verified alumni mentorship onto a single platform.

VI. CONCLUSION

This paper presented KNOTS, an integrated campus networking, placement automation, and institutional governance platform developed using modern full-stack web technologies. The system offers safe, role-based access and centralized data management by combining applicant registration, academic profiling, automated eligibility filtering, 1-click application tracking, profile-driven resume generation, alumni mentorship, and audit logging into a single platform. The evaluation confirms that campus workflows become more transparent, more efficient, and require substantially less manual labor. Future work will expand into federated cross-institutional networks and automated technical assessment screening.

VII. ACKNOWLEDGMENT

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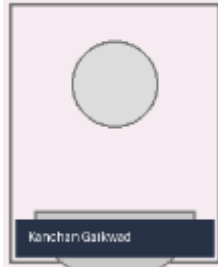
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